

AWS Trusted Advisor

Trusted Advisor provides real-time guidance to help you provision your resources following AWS best practices.

AWS Inspector

Inspector works with EC2 instances to uncover and report vulnerabilities.

AWS Auto Scaling

AWS Auto Scaling will ensure you have the optimal number of EC2 instances to handle your application's load, based on rules you specify. The other services mentioned can help distribute load amongst existing resources, but they do not have the ability by themselves to create new resources.

Auto Scaling Group

The Auto Scaling group can be used to scale out and scale in the instances as the demand dictates. This will save money and avoid having instances sitting idle for long periods of time. AWS Auto Scaling monitors your applications and automatically adjusts your capacity to maintain steady, predictable performance at the lowest possible cost. Using AWS Auto Scaling, it's easy to set up application scaling for multiple resources across multiple services in minutes.

Use Reserved Instances instead of On-Demand Instances.

Reserved Instances are used for longer-term contracts and provide cost savings in these situations. Nothing in this scenario indicates the need for Reserved Instances.

Highly available

Highly available systems are designed to operate continuously without failure for a long time. These systems avoid loss of service by reducing or managing failures.

Durable

Durability is all about long-term data protection. This means your data will remain

intact without corruption.

Elasticity

With elasticity, the company doesn't have to plan ahead of time how much capacity they'll need - elasticity allows them to match the supply of resources with changing workload demands.

Agility

The AWS Cloud provides instant access to new technologies. Companies can move with agility to satisfy new business requirements and meet competitive demands. There is a very low barrier of entry for innovation. If a solution is not meeting expectations, services can be instantly de-provisioned. The other 3 options will also prove to be benefits of deploying in the AWS Cloud, but the use case emphasizes the need to move quickly against competitive threats.

Scalability

This is an example of scalability, which means systems are expected to grow over time with no drop in performance.

Automation

Infrastructure as Code is a key implementation of automation in cloud - using Infrastructure as Code allows you to quickly and easily deploy and manage your environment without reliance on humans to complete all the tasks.

Application Load Balancer

An Application Load Balancer automatically distributes incoming application traffic across multiple EC2 instances based on application variables like content type, user location, and more. The company in this case only has 1 EC2 instance.

Horizontal scaling

Horizontal scaling (or scaling out) adds or replaces EC2 instances automatically across AZs, based on need and changing demand, and is used in the design of

systems with high availability.

Vertical scaling

Vertical scaling is increasing the size and computing power of a single instance or node without increasing the number of nodes or instances.

Fargate

Serverless is a way to build and run applications without having to manage infrastructure. Fargate is considered serverless.

S3

Serverless is a way to build and run applications without having to manage infrastructure. S3 is considered serverless.

DynamoDB

Serverless is a way to build and run applications without having to manage infrastructure. DynamoDB is considered serverless.

DynamoDB is a fully managed NoSQL key-value and document database. DynamoDB is a database technology, but it is not directly compatible with MySQL. DynamoDB is a fast and flexible NoSQL database. NoSQL databases are non-relational and have a dynamic structure, whereas SQL is relational and has a pre-defined structure.

Lambda

Serverless is a way to build and run applications without having to manage infrastructure. Lambda is considered serverless.

EC2

EC2 is not serverless because there is a server that you directly manage.

Elastic Load Balancing

Elastic load balancers are used to balance traffic between EC2 instances, typically in an Auto Scaling group.

Distributing your incoming application traffic across multiple EC2 instances.

CloudWatch Alarms

A CloudWatch alarm can be set up to monitor CPU utilization and trigger further action. Further action could be an Auto Scaling group adding another EC2 instance and/or using SNS to notify team members of the occurrence.

CloudFront

DDoS protection via Shield Advanced is supported on several services, including CloudFront.

CloudFront functions as Amazon's content delivery network, which is a system of distributed servers for delivering web content to users.

Availability Zone

An Availability Zone (AZ) is 1 or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. Each Availability Zone is isolated, but the Availability Zones in a Region are connected through low-latency links.

Edge location

An edge location uses cached copies of your content for fast delivery to users. Don't forget CloudFront speeds up delivery using edge locations. Edge locations consist of over 200+ points of presence around the world that provide fast entry into Amazon's global network. Because of how widespread they are, users can connect to their nearest edge location and have their traffic sent through Amazon's fast global network to reach the resources it needs sooner. This is how CloudFront's caching mechanism works. There are far less Availability Zones and Regions than edge locations, which is why CloudFront does not use them for caching content.

Direct Connect

Direct Connect is a private (bypasses the public internet), dedicated physical network connection from your on-premises data center to AWS. Since the connection is private, it is extremely fast.

Region

A Region is a geographical area of the world that is a collection of data centers logically grouped into Availability Zones and does not cache content.

Secrets Manager

Secrets Manager allows you to manage and retrieve secrets (passwords or keys).

Key Management Service (KMS)

KMS allows you to generate and store encryption keys.

Inspector

Inspector works with EC2 instances to uncover and report vulnerabilities.

Macie

Macie uses machine learning to discover sensitive data stored on Amazon S3. Macie automatically detects a large and growing list of sensitive data types, including personally identifiable information (PII) such as names, addresses, and credit card numbers.

GuardDuty

While GuardDuty has built-in detection for Amazon S3, it only uncovers unauthorized behavior, not personally identifiable information (PII).

Config

Config allows you to assess, audit, and evaluate the configurations of your resources over time. Config works with EC2 instances, servers running on-

premises, and servers and VMs in environments provided by other cloud providers.

Service control policies (SCPs)

AWS Organizations provides central governance and management for multiple accounts. Organization SCPs allow you to create permissions guardrails that apply to all accounts within a given organization.

IAM user policy

While a policy can restrict IAM users from accessing certain services and making changes, this is not the most efficient way to establish controls that all IAM users within an organization adhere to.

AWS VPC

VPC is a foundational service that allows you to create a secure private network in the AWS Cloud to launch your resources.

VPC does not provide automatic replication across AZs.

AWS is responsible for protecting the infrastructure that runs its services. Such services include Amazon VPC, which enables a customer to provision a logically isolated section of the AWS Cloud. This section is simply referred to as a Virtual Private Cloud, or VPC. So, while AWS is responsible for the software (Amazon VPC), the customer is responsible for the virtual network created with that software (VPC), as well as whatever goes in it. This includes subnets (each of which consist of a range of IP addresses) and security groups (each of which acts as a virtual firewall to control inbound and outbound traffic).

AWS Route 53

Route 53 is a DNS service that routes users to applications. Amazon Route 53 effectively connects user requests to infrastructure running in AWS (e.g., Amazon EC2 instances, Elastic Load Balancing load balancers, or Amazon S3 buckets) and can also be used to route users to infrastructure outside of AWS. DDoS protection via Shield Advanced is supported on several services, including Route 53.

Performance Efficiency

This Performance Efficiency pillar focuses on the effective use of resources to

meet demand. In this pillar, you would use the information gathered through the evaluation process to actively drive adoption of new services or resources. You would also define a process to improve workload performance, and you would need to stay up-to-date on new resources and services.

Operational Excellence

The Operational Excellence pillar focus on building applications that effectively support your workloads.

Cost Optimization

The Cost Optimization pillar focuses on building resilient systems at the least cost.

IAM credential report

The IAM credential report lists all the users and the status of their various credentials, including passwords, access keys, server certificates, and MFA devices.

Identity and Access Management (IAM)

IAM allows you to create and manage access keys for an IAM user.

Bucket access policy

A bucket access policy can be attached directly to an S3 bucket to limit access to specific users.

Bucket policies

Bucket policies allow you to control access to entire buckets, whereas access control lists let you control access to individual objects within an S3 bucket.

CloudWatch

CloudWatch can monitor the state of your AWS resources and can notify you when

an EC2 is approaching 100% utilization.

CloudWatch Alarms can be used to determine the percentage of utilization versus the limit.

Database Migration Service

DMS supports homogeneous migrations like Oracle to Oracle and heterogeneous migrations like Oracle to Aurora, with minimal downtime.

Snowmobile

While Snowmobile is an option to transfer data to AWS, it supports multi-petabyte or exabyte-scale data and doesn't convert from one database engine to another.

AWS Kinesis

Kinesis allows you to analyze data and video streams in real time.

AWS Personal Health Dashboard

The AWS Personal Health Dashboard publishes alerts and remediation guidance when issues with AWS services arise. Notifications are also provided for scheduled events that may impact AWS customers.

AWS Partner Network (APN) Consulting Partners

APN Consulting Partners include professional services organizations like system integrators, strategic consultancies, agencies, managed service providers (MSPs), and value-added resellers. In this case, we would engage a Consulting Partner to help us deploy a new system to the AWS Cloud.

Lambda

Lambda allows you to run a variable and intermittent code without paying for compute time when your code isn't running.

Lightsail

Lightsail allows you to quickly launch all the resources you need for small projects.

Amazon Redshift

Amazon Redshift provides the best solution for performing queries based on a predefined set of dimensions. Redshift organizes data for high performance based on user-specified distribution schemes. Amazon ElastiCache provides in-memory performance, but no data organization assistance. Amazon Aurora and Amazon DynamoDB are good solutions, but Redshift's columnar storage gives it the edge. Redshift is a scalable data warehouse solution that supports querying, reporting, analytics, and business intelligence. It can be used when you need to consolidate multiple data sources for reporting and don't require real-time transaction processing (insert, update, and delete).

Redshift is AWS' data warehousing service

Amazon ElastiCache

ElastiCache is a fully managed in-memory datastore compatible with Redis or Memcached.

Utilize loosely coupled components

Loosely coupled components are connected but not dependent on each other. Loose coupling reduces the risk of cascading failures and can be achieved through asynchronous integration.

Design the architecture to reduce inter-dependencies

Architectures should be designed in a way that reduces inter-dependencies and the risk of cascading failures between components.

Translate

Translate provides language translation.

SageMaker

SageMaker helps you build, train, and deploy machine learning models quickly

S3 Glacier Deep Archive

S3 Glacier Deep Archive is used for long-term data archival that only needs to be accessed once or twice a year. It is often used to store data for regulatory compliance. If faster access times are needed, there is also S3 Glacier.

Rekognition

The company can use Rekognition to identify objects like protective equipment in their images and detect if employees are wearing the required protective equipment.

S3 One Zone-Infrequent Access

S3 One Zone-Infrequent Access is designed for customers who want a lower cost option for infrequently accessed data but do not require the multiple Availability Zone data resilience model of the S3 Standard or S3 Standard-Infrequent Access storage classes. S3 One Zone-Infrequent Access provides millisecond access when the data is needed.

Snowmobile

The Snow Family allows you to transfer large amounts of on-premises data to AWS using a physical device. Snowmobile transports multi-petabyte or exabyte-scale data.

AWS Snowball

Snowball helps you migrate massive amounts of data into cloud, so it is considered a migration tool.

Systems Manager

Systems Manager gives you visibility and control over your AWS resources.

Tags

The administrator can use tags to identify resources with the Marketing department.

Enterprise Support plan

Enterprise Support provides access to a Technical Account Manager (TAM) who helps coordinate access to subject matter experts among other things. The response time for a business-critical system down case is less than 15 minutes. AWS Infrastructure Event Management is a structured program available to Enterprise Support customers (and Business Support customers for an additional fee) that helps you plan for large-scale events, such as product or application launches, infrastructure migrations, and marketing events.

Business Support plan

The Business level support plan provides 1 hour or less response time support for production-level failures. Also is the minimum plan that provides access to the full set of Trusted Advisor checks.

Organizations

Organizations allows you to centrally manage multiple AWS accounts under 1 umbrella. You can allocate resources and apply policies across accounts.

Control Tower

Control Tower helps you ensure your accounts conform to company-wide policies. Control Tower actually sits on top of Organizations.

AWS Cost Explorer

Cost Explorer allows you to visualize and forecast your costs and usage over time.

AWS Pricing Calculator

The Pricing Calculator provides an estimate of AWS fees and charges. Since the company knows the workload details, the AWS Pricing Calculator can also help with calculating the total cost of ownership.

AWS Artifact

AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to AWS' security and compliance reports and select online agreements. AWS Support's site provides general information about compliance, but it does not hold the secure reports or certificates. AWS IAM is used to control security within your AWS Account. Amazon Macie is a data security classification service.

Amazon WorkSpaces

Amazon WorkSpaces is a managed, secure Desktop-as-a-Service (DaaS) solution. You can use Amazon WorkSpaces to provision either Windows or Linux desktops in just a few minutes and quickly scale to provide thousands of desktops to workers across the globe.

Private cloud

With the private cloud, all resources run on-premises. Don't forget: This is your internal data center.

Hybrid cloud

With the hybrid cloud, some resources run in the cloud while others run on-premises. This means resources run in the AWS Cloud and your internal data center.

Customer data

Customers are responsible for the storage and securing of their own data.

AWS

The platform-as-a-service model removes the need for organizations to manage the underlying infrastructure (usually hardware and operating systems) and allows you to focus on the deployment and management of your applications.

AWS Shield

Shield is a managed Distributed Denial of Service (DDoS) protection service. Shield Standard provides free protection against common and frequently occurring attacks. Shield Advanced provides enhanced protections and 24/7 access to AWS experts for a fee.

Web Application Firewall (WAF)

There is a rule type in WAF called a "rate-based" rule that protects you from web-layer DDoS attacks, brute-force login attempts, and bots. (Note: This requires very detailed knowledge of WAF and will probably not be on the exam, but it's good to know just in case.)

WAF helps protect your web applications against common web attacks like SQL injection attacks and cross-site scripting.

Security groups

Security groups act like built-in firewalls for your virtual servers — the rules you create define what is allowed to talk to your instances and how. Although network access control lists can be used to block or deny traffic, these operate at the subnet level (covering all instances in the subnet with the same ruleset), not per instance as the question specifies. Route tables tell traffic where it should go next to reach its destination, and an Availability Zone is a collection of data centers — which isn't relevant in this question.

Subnets

AWS is responsible for protecting the infrastructure that runs its services. Such services include Amazon VPC, which enables a customer to provision a logically isolated section of the AWS Cloud. This section is simply referred to as a Virtual Private Cloud, or VPC. So, while AWS is responsible for the software (Amazon VPC), the customer is responsible for the virtual network created with that

software (VPC), as well as whatever goes in it. This includes subnets (each of which consist of a range of IP addresses) and security groups (each of which acts as a virtual firewall to control inbound and outbound traffic).

Authentication

Authentication defines the "who" in regard to access. Authentication is where you present your identity (username) and provide verification (password).

Cost Optimization

The Cost Optimization pillar focuses on building resilient systems at the least cost.

Operational Excellence

The Operational Excellence pillar focuses on building applications that effectively support your workloads.

NAT gateway

A NAT gateway is required to allow resources in a private subnet to access the internet.

Elastic Beanstalk

Elastic Beanstalk allows you to deploy your web applications and web services to AWS.

```
Load Balancers {  
    Network  
    Application  
    Classic  
}
```

Aurora

Aurora is AWS' managed database service that is up to 5x faster than a traditional MySQL database.

Aurora is a type of RDS engine on AWS.

Aurora is a relational database compatible with MySQL and PostgreSQL that was created by AWS.

MariaDB

MariaDB is a type of RDS engine on AWS.

CloudFormation

CloudFormation provides the ability to provision a repeatedly deployable environment for your customers.

AWS CloudFormation is a service that helps you model and set up your Amazon Web Services resources so you can spend less time managing those resources and more time focusing on your applications that run in AWS.

CloudTrail

CloudTrail tracks user activity and API calls within your account.

CloudTrail tracks the AWS Region that the request was made to, such as us-east-1.

CloudTrail allows you to track the username.

Simple Notification Service (SNS)

With Simple Notification Service, or Amazon SNS, you can create and appropriately name a topic to which you publish your messages. Then, the select Tech Support members can subscribe to that topic to receive these messages as email notifications.

Simple Email Service (SES)

SES is an email service that allows you to send richly formatted HTML emails from your applications. SES is ideal for marketing campaigns, not the scenario described.

Submit a service limit increase.

The key word here is "quickest," and the fastest option is to request a service limit increase at the AWS Support Center.

Lex

Lex helps you build conversational interfaces like chatbots.

Network Load Balancer

Network Load Balancer is best suited for load balancing of Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Transport Layer Security (TLS) traffic where extreme performance is required. Operating at the connection level (Layer 4), Network Load Balancer routes traffic to targets within Amazon Virtual Private Cloud (Amazon VPC) and is capable of handling millions of requests per second while maintaining ultra-low latencies.

Application Load Balancer

An Application Load Balancer is best suited for load balancing of HTTP and HTTPS traffic and provides advanced request routing targeted at the delivery of modern application architectures, including microservices and containers.

CodeCommit

CodeCommit is a source control system for private Git repositories.

AWS Marketplace

Marketplace is a digital catalog of prebuilt solutions you can purchase or license. You may also use it to sell solutions to others.

Marketplace is a digital catalog of prebuilt solutions you can purchase or license. You may also sell your own solutions to others via Marketplace.

Managed Services

Managed Services helps you efficiently operate your AWS infrastructure and

reduces operational risks and overhead.

Professional Services

Professional Services helps enterprise customers move to a cloud-based operating model.

AWS Infrastructure Event Management

Infrastructure Event Management offers architecture guidance and operational support during the preparation and execution of planned events, such as shopping holidays, product launches, and migrations.

Standard Reserved Instances

Standard Reserved Instances provide you with a significant discount (up to 72%) compared to On-Demand Instance pricing and can be purchased for a 1-year or 3-year term.

Public cloud

With the public cloud, all resources run in the cloud. Don't forget: This is the AWS Cloud.

AWS SQS

SQS is a message queuing service that allows you to build loosely coupled systems.

Key Management Service (KMS)

When you create an encrypted Amazon EBS volume, you're able to specify a KMS customer master key.

Customer Service Policy for Penetration Testing

AWS customers are welcome to carry out security assessments or penetration

tests against their AWS infrastructure without prior approval for Amazon EC2 instances, NAT gateways, elastic load balancers, and 7 other services.

Service control policies

AWS Organizations provides central governance and management for multiple accounts. Organization service control policies (SCPs) allow you to create permissions guardrails that apply to all accounts within a given organization.

Cloud9

Cloud9 allows application developers to write code within an integrated development environment (IDE) from within their web browser.

Internet gateway

An internet gateway enables resources inside your VPC to reach the internet, as long as route tables and IP addresses are correctly configured in your environment.

An internet gateway allows public traffic to the internet from the VPC.

Global Accelerator

Global Accelerator can improve the experience by routing player traffic along with the private AWS global network to the fastest instance of your application. Player traffic is not negatively impacted by internet congestion and local outages.

ElastiCache

ElastiCache helps you alleviate database load for data that is accessed often. ElastiCache is a great way to cache common queries.

Spot Instance

Spot Instances are a great choice for this use case. Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS Cloud. Spot Instances are available at up to a 90% discount compared to On-Demand prices. The key phrase in this question is, "It is alright if there are interruptions in the application."

If the application could not accept interruptions, then the best option would be On-Demand.

On-Demand

On-Demand is where you pay for the resources you have used at the end of the billing period.

Reserved

Reserved is where you commit to paying a certain amount every billing period for a length of time, no matter if you used them or not.

EC2 Reserved Instances offer significant discounts for a contracted term-of-service, up to 75% off.

Billing support

The AWS Support Basic tier is the free support given to all AWS accounts. All accounts receive billing support, because every customer needs an avenue to lodge disputes or make requests around their billing.